Generated by Scopus AI, Sun Oct 11 2026

Intelligent Dispatching for Railways

RailSync: Intelligent Dispatching and Real-Time Schedule Optimization for Modern Railways

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
System Architectures & Multi-Agent Integration	Network-centric, multi-agent, and AI-driven architectures enable scalable, adaptive, and fault-tolerant dispatching.	1 2 3 4 5
Hybrid Optimization Algorithms	Combining MILP with metaheuristics (e.g., tabu search, genetic algorithms) balances solution quality and speed.	6 7 8 9
Data Integration & Real-Time Fusion	Advanced fusion (e.g., adaptive Kalman filters, knowledge graphs) supports precise, conflict-free scheduling.	10 11 12 13
Case Studies & Scalability	RailSync-like systems improve efficiency but face challenges in scaling, legacy integration, and real-world complexity.	14 15 16 17
AI Integration & Human Collaboration	AI-driven simulation, reinforcement learning, and LLMs are increasingly blended with human-centric workflows.	18 19 20 21

Direct Answer

RailSync and similar modern railway dispatching systems employ advanced multi-agent architectures, hybrid optimization algorithms, and sophisticated real-time data fusion techniques. These systems are designed to handle dynamic disturbances and scale efficiently through the integration of incremental digitalization with legacy systems, thereby improving train scheduling, safety, and operational costs.

Study Scope

- **Time Period:** Focus on advancements and case studies from 2020 onward, with foundational references to earlier work.
- **Disciplines:** Operations research, artificial intelligence, transportation engineering, computer science, and

systems integration.

- **Methods:** Meta-analysis of empirical studies, case studies, algorithmic benchmarking, and system architecture reviews.

Assumptions & Limitations

- **Assumptions:** The report assumes the availability of high-quality, multi-source operational data and the feasibility of integrating AI with legacy systems.
- **Limitations:** Real-world deployment is often constrained by legacy infrastructure, data quality, and the gap between theoretical models and operational requirements. Scalability and interpretability of AI in safety-critical contexts remain open challenges.

Suggested Further Research

- Integration of digital twins with multi-agent reinforcement learning for virtualized testing of dispatching strategies.
- Development of knowledge graph-based interoperability frameworks for cross-border railway operations.
- Enhanced AI interpretability and human-AI collaboration models for safety-critical decision-making.
- Scalable, distributed optimization methods for ultra-large networks.

1. Introduction

Overview of Intelligent Railway Dispatching

Modern railway operations are undergoing a paradigm shift from legacy, analog-based dispatching to intelligent, digital, and AI-driven systems. Intelligent dispatching and real-time schedule optimization are central to this transformation, leveraging advanced system architectures, optimization algorithms, and integrated data sources to enhance operational efficiency, safety, and adaptability. The transition is motivated by the need to manage increasing traffic density, complex network topologies, and dynamic disturbances, all while integrating with existing infrastructure and ensuring safety-critical reliability [1] [15].

Synthesis:
The evolution from manual, regionally siloed dispatching to network-wide, data-driven, and AI-augmented platforms marks a significant leap in railway operational management. This shift is characterized by the adoption of multi-agent architectures, hybrid optimization frameworks, and real-time data fusion, collectively enabling more responsive, scalable, and resilient railway systems.

2. Theoretical Frameworks

System Architectures for Intelligent Dispatching

Network-Centric and Multi-Agent Architectures

- **Network-centric multi-agent systems** decompose large railway networks into infrastructure-defined control units, each managed by intelligent agents. This modularity enhances scalability, computational efficiency, and fault tolerance [1] [2] [3] [22].
- **Multi-agent deep reinforcement learning (MADRL)** frameworks enable agents to collaboratively schedule and reschedule trains, handling complex spatio-temporal constraints and disruptions more efficiently than traditional methods [2] [3] [23].
- **MAPF-based architectures** (Multi-Agent Path Finding) use prioritized planning and parallel computing to plan collision-free paths for thousands of trains in real time, supporting deadlock-free dispatching [24] [25].
- **Emerging integration with large language models (LLMs)** and agentic AI systems offers modular, scalable, and adaptive solutions for complex railway tasks, including data governance and operational management [4] [26] [27].

Hybrid Optimization and Decision Support Systems

- **Hybrid frameworks** combine exact methods (e.g., MILP) with metaheuristics (e.g., tabu search, genetic algorithms) to balance solution quality and computational speed, especially under dynamic disturbances [14] [28] [29] [30] [31].
- **Decision support systems** like AGLIBRARY operate at microscopic levels, producing conflict-free schedules with second-level precision and minimizing delays and deadlocks [14].
- **Reinforcement-inspired feedback mechanisms** and adaptive priority weighting further enhance performance in complex, real-time environments [29].

AI-Driven Centralized Dispatcher Control

- **AI-driven centralized control systems** automate train movement plotting, fault prediction, delay reduction, and energy efficiency, enhancing safety and operational productivity [15] [18] [32] [33].
- **Migration strategies** involve staged, iterative integration with legacy infrastructure, preserving existing equipment while introducing digital dispatching centers and IP-based networks [15].
- **Human-AI collaboration** is critical, requiring transparency, trust, and explainability to ensure effective integration in safety-critical contexts [18].

Digitalization and Software-Centric Platforms

- **Digitalization** involves migrating from analog to software-centric platforms, enabling modular scalability,

real-time supervision, automated diagnostics, and improved maintainability [15] [34].

- **Software-centric platforms** facilitate centralized monitoring, regulatory compliance, and future automation, supporting seamless integration with intelligent multi-agent dispatching architectures [15].

Synthesis:
The convergence of network-centric, multi-agent, and AI-driven architectures with hybrid optimization and digitalization strategies forms the backbone of modern intelligent dispatching systems. These frameworks enable scalable, adaptive, and resilient railway operations, capable of integrating with legacy infrastructure and supporting real-time, safety-critical decision-making.

3. Methods & Data Transparency

Algorithms for Real-Time Schedule Optimization

Metaheuristic and Bioinspired Algorithms

- **Variable neighbourhood search and tabu search** are effective for real-time train scheduling, providing conflict-free schedules and minimizing delays, often outperforming commercial solvers in disturbed scenarios [6] [30] [35] [36] [37].

- **Ant colony optimization (ACO) and genetic algorithms (GA)** excel in complex, dynamic environments, reducing delay costs by up to 30% compared to simpler methods [6] [37].

- **Hybrid metaheuristics** (e.g., Bat Algorithm, strengthened elitist genetic algorithms) and reinforcement learning approaches further enhance convergence speed and solution quality under dynamic disturbances [19] [36].

Mixed Integer Linear Programming and Decomposition

- **MILP models with decomposition** (e.g., Lagrangian relaxation, column generation, Benders decomposition) are widely used for operational dispatching, enabling optimal or near-optimal solutions for large-scale, complex constraints [38] [39] [40] [41] [42].

- **Decomposition approaches** partition problems into manageable subproblems (e.g., line and station levels), improving computational efficiency and scalability [39] [43].

Hybrid Optimization Frameworks

- **Hybrid frameworks** combine MILP and metaheuristics, decomposing problems by variable type and leveraging parallelism to balance solution quality and computational speed [8] [30] [44] [45].

- **Logic-based Benders decomposition** and reinforcement learning integration further accelerate computation and improve solution quality [41] [44].

Rolling Horizon and Closed-Loop Scheduling

- **Rolling horizon frameworks** dynamically adjust schedules in response to stochastic disruptions, enabling adaptive recovery strategies and improved operational efficiency [46] [47].
- **Closed-loop frameworks** integrate simulation and optimization, allowing real-time feedback and continuous schedule refinement [48] [49].

Case Example: GE-Norfolk Southern System

- The **GE-Norfolk Southern real-time scheduling system** increased average train speeds by two miles per hour, saving $200 million annually and improving schedule adherence through network-wide, algorithm-based dispatching [50].

Synthesis:
The integration of metaheuristics, MILP, and hybrid frameworks enables robust, real-time schedule optimization capable of handling dynamic disturbances and large-scale operational complexity. Rolling horizon and closed-loop approaches further enhance adaptability and resilience.

4. Data Integration and Real-Time Information Sources

Types of Data Sources and Integration Methods

- **Data sources:** Sensor networks, operational data, environmental data, GPS, GSM, RFID, and big data platforms [51] [52] [53] [54] [55].
- **Integration technologies:** ICT, AI, big data analytics, and cloud computing support real-time data collection, processing, and decision support [53] [56] [57].

Real-Time Data Fusion Techniques

- **Adaptive Kalman filters** and kernel-based smoothing integrate heterogeneous spatiotemporal data for improved prediction and continuous monitoring [10] [12] [58].
- **Knowledge graphs and semantic technologies** enhance interoperability and analytical capabilities, especially in cross-border operations [59] [60] [61].
- **Multi-source heterogeneous data fusion** at data, feature, and decision levels supports operational intelligence and predictive maintenance [60] [62].

Integrated Dispatching Assistance Systems

- **Real-time dispatching assistance systems** combine train location detection, delay monitoring, and automated conflict resolution to support flexible, automated operations [63] [64] [65].
- **Machine learning models** (e.g., LSTM) enhance arrival time prediction and operational decision support [65].

Criteria for Real-Time Information Source Selection

- **Quality factors:** Information availability, reliability, user requirements, and system performance indicators guide source selection [66] [67] [68].
- **Evaluation frameworks:** Multi-criteria decision-making and expert validation ensure system reliability and certification [67] [68].

Microscopic-Level Scheduling and Precision

- **Second-level precision** is achieved through MILP models, path-oriented binary decision models, and decomposition approaches, enabling conflict-free operation and rapid rescheduling [14] [69] [70] [71].

Synthesis:
Effective real-time schedule optimization relies on the integration and fusion of heterogeneous data sources. Advanced fusion techniques, semantic technologies, and robust evaluation frameworks ensure high-quality, reliable, and actionable information for intelligent dispatching.

5. Critical Analysis of Findings

Case Studies and Implementation Challenges

Key Findings from RailSync-Like Case Studies

- **Operational benefits:** Near-optimal, conflict-free scheduling, reduced delays, and improved safety and efficiency [14] [28] [32] [50] [72].
- **Scalability challenges:** High computational complexity in dense networks, requiring decomposition, distributed optimization, and metaheuristics for timely solutions [72] [73].
- **Adaptability and integration:** Incremental digitalization strategies enable seamless migration from legacy analog to software-centric platforms, preserving operational continuity [15] [21] [74].
- **Decomposition approaches:** Master-slave algorithms and Benders-like decomposition have proven effective in operational systems (e.g., Norway), improving scalability and solution quality [17] [38] [71] [75].
- **Theory-practice gap:** Practical deployment is hindered by data quality, system modeling challenges, and alignment with user requirements [16] [38] [76] [77] [78].

Scalability Challenges in Dense Railway Networks

- **Decomposition and distributed optimization** (e.g., geography-based, train-based, time-interval-based) are essential for handling large-scale, dense networks [73].
- **Multi-agent systems** and collaborative voting mechanisms improve scalability and solution quality by reducing search space and avoiding localized optimizations [79].

- **Integration of scheduling and control** is critical for maintaining punctuality and operational efficiency [80].

Incremental Digitalization and Migration Strategies

- **Hybrid migration architectures** integrate digital and analog components, enabling modular scalability, real-time supervision, and improved maintainability [15] [34].
- **Cybersecurity** and workforce training are essential to address new vulnerabilities introduced by digitalization [21] [81].

Gaps Between Theory and Practice

- **Limited real-world deployment:** Despite advances in simulation and optimization, practical implementations remain scarce due to trust, data quality, and operational alignment issues [16] [76].
- **Integration challenges:** Bridging microscopic and macroscopic models, and aligning theoretical assumptions with operational realities, are ongoing challenges [77] [82] [83].

Synthesis:
While RailSync-like systems demonstrate significant operational benefits, real-world deployment is constrained by scalability, integration with legacy infrastructure, and the gap between theoretical models and practical requirements. Incremental digitalization, decomposition strategies, and robust human-AI collaboration frameworks are critical for overcoming these challenges.

6. Real-World Implications

Operational Efficiency and Cost Savings

- **Case studies** (e.g., GE-Norfolk Southern) show substantial improvements in train speed, schedule adherence, and cost savings through algorithm-based, network-wide dispatching [50].
- **Automated conflict resolution** and real-time rescheduling reduce delays and enhance safety, supporting more reliable and efficient railway operations [14] [28].

Scalability and Adaptability

- **Decomposition and distributed optimization** enable real-time scheduling in large, dense networks, supporting scalability and rapid response to disturbances [73].
- **Incremental digitalization** ensures seamless migration and integration with legacy systems, preserving operational continuity and enabling future automation [15] [34].

Human-AI Collaboration and Trust

- **Human-AI collaboration frameworks** are essential for safety-critical operations, ensuring transparency,

trust, and effective decision-making [16] [18] [84].

- **AI-driven simulation and reinforcement learning** augment human expertise, supporting adaptive and resilient operational management [19] [20].

Data-Driven Decision Support

- **Advanced data fusion** and real-time information integration underpin precise, conflict-free scheduling and predictive maintenance, enhancing operational intelligence [10] [11] [12] [13].

- **Knowledge graph-based interoperability** supports cross-border operations and seamless data sharing between disparate systems [59] [61].

Synthesis:
The practical impact of RailSync-like systems is evident in improved operational efficiency, cost savings, and enhanced safety. Scalability, adaptability, and robust human-AI collaboration are critical for realizing these benefits in real-world, large-scale railway networks.

7. Future Research Directions

Emerging Technologies and Architectural Innovations

- **Digital twins** integrated with multi-agent deep reinforcement learning offer virtual sandboxes for testing real-time dispatching strategies [4] [26] [27] [85].

- **Cloud-based platforms** and microservice orchestration enable flexible, scalable management of railway operations [86].

- **Multimodal large model technologies** address sector-specific challenges, supporting adaptive and intelligent operational management [27].

Scalability and Computational Trade-Offs

- **MAPF-based and multi-agent architectures** face computational limits (NP-hardness) in ultra-large networks, necessitating trade-offs between optimality, speed, and scalability [87] [88] [89] [90].

- **Distributed learning and hierarchical control** improve scalability and coordination in large-scale, real-time dispatching [91] [92].

Integration and Interoperability Challenges

- **Knowledge graph-based data fusion** and semantic technologies enhance interoperability across heterogeneous data sources and cross-border operations [20] [61] [93].

- **Standardization and regulatory alignment** are needed to overcome fragmented adoption and ensure

seamless integration [20].

Human-AI Collaboration and Trust

- **Transparency, explainability, and trust** are paramount for human-AI collaboration in safety-critical railway dispatching [16] [18] [84].
- **Blended human-AI frameworks** support robust, adaptive decision-making and fault diagnosis [18] [19] [20] [21].

Synthesis:
Future research should focus on integrating digital twins, scalable distributed optimization, knowledge graph-based interoperability, and enhanced human-AI collaboration to address the remaining gaps in scalability, adaptability, and real-world deployment of intelligent railway dispatching systems.

Conclusion

RailSync and similar intelligent dispatching systems represent the forefront of modern railway operational management, combining advanced multi-agent architectures, hybrid optimization algorithms, and sophisticated real-time data fusion. While significant progress has been made in improving operational efficiency, safety, and scalability, challenges remain in integrating with legacy infrastructure, scaling to ultra-large networks, and bridging the gap between theoretical models and practical deployment. Continued research and innovation in digital twins, distributed optimization, semantic interoperability, and human-AI collaboration are essential for realizing the full potential of intelligent, adaptive, and resilient railway systems.

References

1. Network-centric approach to real-time train scheduling in large-scale railway systems Belousov, A.A., Skobelev, P.O., Stepanov, M.E. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2015 https://www.scopus.com/pages/publications/84947766102?origin=scopusAI

2. Causal reinforcement learning for train scheduling on single-track railway networks Yang, F., Liu, J., Li, J., (...), Ni, S. Transportation Research Part C: Emerging Technologies, 2025 https://www.scopus.com/pages/publications/105008560856?origin=scopusAI

3. Adaptive rescheduling of rail transit services with short-turnings under disruptions via a multi-agent deep reinforcement learning approach Ying, C., Chow, A.H.F., Yan, Y., (...), Wang, S. Transportation Research Part B: Methodological, 2024 https://www.scopus.com/pages/publications/85203005968?origin=scopusAI

4. Architecture Design of Multi-Agent LLM Systems in Railway Data Governance Cui, Y., Guo, Y., Ren, C., (...), Yin, Z. Proceedings - 2025 27th IEEE International Conference on High Performance Computing and Communications, 11th IEEE International Conference on Data Science and Systems, 23rd IEEE International Conference on Smart City, 11th IEEE International Conference on Dependability in Sensor, Cloud, and Big Data Systems and Applications and 21st IEEE International Conference on Embedded Software and Systems, HPCC/DSS/SmartCity/DependSys/ICESS 2025, 2025 https://www.scopus.com/pages/publications/105022704383?origin=scopusAI

5. An agent-based support system for railway station dispatching Zhu, T., Mera, J.M., Suarez, B., Maroto, J. Expert Systems with Applications, 2016 https://www.scopus.com/pages/publications/84971283614?origin=scopusAI

6. A comparison of algorithms for minimising delay costs in disturbed railway traffic scenarios Fan, B., Roberts, C., Weston, P. Journal of Rail Transport Planning and Management, 2012 https://www.scopus.com/pages/publications/84870060742?origin=scopusAI

7. An efficient evolutionary algorithm for high-speed train rescheduling under a partial station blockage Wang, R., Zhang, Q., Dai, X., (...), Jin, Y. Applied Soft Computing, 2023 https://www.scopus.com/pages/publications/85164719172?origin=scopusAI

8. A parallel algorithm for matheuristics: A comparison of optimization solvers González, M., López-Espín, J.J., Aparicio, J. Electronics (Switzerland), 2020 https://www.scopus.com/pages/publications/85091642388?origin=scopusAI

9. Metaheuristics for Real-Time Near-Optimal Train Scheduling and Routing Sama, M., Dariano, A., Toli, A., (...), Corman, F. IEEE Conference on Intelligent Transportation Systems, Proceedings, ITSC, 2015 https://www.scopus.com/pages/publications/84950284995?origin=scopusAI

10. Modeling and data fusion of dynamic highway traffic Ottaviano, F., Cui, F., Chow, A.H.F. Transportation Research Record, 2017 https://www.scopus.com/pages/publications/85027543693?origin=scopusAI

11. Integrated Sensing-Positioning-Communication and Application in Smart Railway Systems Liu, G., Jiang, H., Zhang, Q., (...), Xiao, M. IEEE Communications Magazine, 2026 https://www.scopus.com/pages/publications/105039318636?origin=scopusAI

12. A robust and adaptive framework for localization under varying sensor modalities Bopardikar, S.D., Zhang, S., Speranzon, A. AIAA Guidance, Navigation, and Control (GNC) Conference, 2013 https://www.scopus.com/pages/publications/85088760621?origin=scopusAI

13. An advanced real-time train dispatching system for minimizing the propagation of delays in a dispatching area under severe disturbances D'Ariano, A., Pranzo, M. Networks and Spatial Economics, 2009 https://www.scopus.com/pages/publications/59749089450?origin=scopusAI

14. Evaluating the applicability of advanced techniques for practical real-time train scheduling D'Ariano, A., Samà, M., D'Ariano, P., Pacciarelli, D. Transportation Research Procedia, 2014 https://www.scopus.com/pages/publications/84942590300?origin=scopusAI

15. Digitalization of Railway Traffic Dispatching Systems: From Legacy Infrastructure to a Software-Centric Platform Kokić, I., Vuleta-Radoičić, J., Salom, I., (...), Rakas, S.B. Computers, 2026 https://www.scopus.com/pages/publications/105034132774?origin=scopusAI

16. A collection of aspects why optimization projects for railway companies could risk not to succeed – A multi-perspective approach Liebchen, C., Schülldorf, H. Journal of Rail Transport Planning and Management, 2019 https://www.scopus.com/pages/publications/85073145756?origin=scopusAI

17. The track formulation for the train dispatching problem Lamorgese, L., Mannino, C. Electronic Notes in Discrete Mathematics, 2013 https://www.scopus.com/pages/publications/84879726563?origin=scopusAI

18. Toward a Holistic Framework for Human-AI Collaboration in Safety-Critical Systems Bessa, R.J., Leyli-Abadi, M., Yagoubi, M., (...), Zanotti, G. Artificial Intelligence, Data and Robotics: Foundations, Transformations and Future Directions, 2026 https://www.scopus.com/pages/publications/105041404647?

origin=scopusAI

19. Reinforcement Learning for Online Dispatching Policy in Real-Time Train Timetable Rescheduling Yue, P., Jin, Y., Dai, X., (...), Cui, D. IEEE Transactions on Intelligent Transportation Systems, 2024 https://www.scopus.com/pages/publications/85168702119?origin=scopusAI

20. Inter-railways data sharing for seamless travelling applications in Europe Sohi, S., Harikrishnan, S., Labanca, G., (...), Polleres, A. CEUR Workshop Proceedings, 2025 https://www.scopus.com/pages/publications/105019642102?origin=scopusAI

21. A review on cybersecurity in railways Kour, R., Patwardhan, A., Thaduri, A., Karim, R. Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit, 2023 https://www.scopus.com/pages/publications/85129455821?origin=scopusAI

22. Transformer-HAPPO: A convergent multi-agent reinforcement learning framework for large-scale single-track railway timetabling Li, W., Ni, S., Lv, M., (...), Zhang, L. Transportation Research Part B: Methodological, 2026 https://www.scopus.com/pages/publications/105048158494?origin=scopusAI

23. Large-Scale Metro Train Timetable Rescheduling via Multi-Agent Deep Reinforcement Learning: A High-Dimensional Optimization Approach in Flatland Environment Yang, J., Yang, H., Wang, W., Xia, C. Applied Sciences (Switzerland), 2026 https://www.scopus.com/pages/publications/105035518960?origin=scopusAI

24. Scalable Rail Planning and Replanning: Winning the 2020 Flatland Challenge Li, J., Chen, Z., Zheng, Y., (...), Koenig, S. 14th International Symposium on Combinatorial Search, SoCS 2021, 2021 https://www.scopus.com/pages/publications/85124599838?origin=scopusAI

25. Scalable Rail Planning and Replanning: Winning the 2020 Flatland Challenge Li, J., Chen, Z., Zheng, Y., (...), Koenig, S. Proceedings International Conference on Automated Planning and Scheduling, ICAPS, 2021 https://www.scopus.com/pages/publications/85105819190?origin=scopusAI

26. Building Agentic AI Systems with LangGraph: From LLM Pipelines to Multi-Agent Decision Workflows Kuhail, M.A. Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2026 https://www.scopus.com/pages/publications/105051093284?origin=scopusAI

27. Multimodal Large Model Technology and Application for Railways Hu, F., Gao, J., Hua, F., (...), Han, W. Proceedings - 2024 International Conference on Computational Linguistics and Natural Language Processing, CLNLP 2024, 2024 https://www.scopus.com/pages/publications/85211333673?origin=scopusAI

28. An optimized, automatic TMS in operations in Roma Tiburtina and Monfalcone stations Foglietta, S., Leo, G., Mannino, C., (...), Piacentini, M. WIT Transactions on the Built Environment, 2014 https://www.scopus.com/pages/publications/84903744956?origin=scopusAI

29. Hybrid Adaptive Heuristic Scheduling Framework for Railway Transportation Organization Optimization Zhao, W., Li, L. International Conference on Communication, Computer and Information Technology, IC3IT 2025, 2025 https://www.scopus.com/pages/publications/105033349739?origin=scopusAI

30. A variable neighbourhood search for fast train scheduling and routing during disturbed railway traffic situations Samà, M., D′Ariano, A., Corman, F., Pacciarelli, D. Computers and Operations Research, 2017 https://www.scopus.com/pages/publications/84959308471?origin=scopusAI

31. Rapid Rescheduling of Cross-Line Train Timetables Under Delay Scenarios: Engineering Practice of a GAPSO Hybrid Algorithm Peng, Z., Mu, H. Lecture Notes in Electrical Engineering, 2026 https://www.scopus.com/pages/publications/105040636948?origin=scopusAI

32. Artificial Intelligence-based Centralized Dispatcher Control System for Organizing Train Movement Boltayev, S.T., Muhiddinov, O.O., Kosimova, Q.A. Proceedings - 2026 International Russian Smart Industry Conference, SmartIndustryCon 2026, 2026 https://www.scopus.com/pages/publications/105040970585?origin=scopusAI

33. Cybersecurity Aspects of Artificial Intelligence Driven System for Railway Infrastructure Telemetry Atanasov, I., Pencheva, E. Conference Proceedings - 2026 IEEE International Workshop on Metrology for Industry 4.0 and IoT, MetroInd4.0 and IoT 2026, 2026 https://www.scopus.com/pages/publications/105048795020?origin=scopusAI

34. A Train to the OPERA Cavada, R., Cimatti, A., Scaglione, G., (...), Tessi, M. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105041190612?origin=scopusAI

35. Intelligent decision making through bioinspired optimization: Ant colony optimization for real-world scenarios Chellam, J.A.I., Muthukumar, P., Selvi, M.S. Intelligent Decision Making Through Bio-Inspired Optimization, 2024 https://www.scopus.com/pages/publications/85192822585?origin=scopusAI

36. Optimal Scheduling for Delay Management in Railway Network Using Hybrid Bat Algorithm Dalapati, P., Paul, K. Lecture Notes in Electrical Engineering, 2021 https://www.scopus.com/pages/publications/85101531171?origin=scopusAI

37. Hybrid ant colony optimization based on genetic algorithm for container loading problem Zhang, D., Du, L. Proceedings of the 2011 International Conference of Soft Computing and Pattern Recognition, SoCPaR 2011, 2011 https://www.scopus.com/pages/publications/83655191036?origin=scopusAI

38. Optimal train dispatching by Benders'-like reformulation Lamorgese, L., Mannino, C., Piacentini, M. Transportation Science, 2016 https://www.scopus.com/pages/publications/84984868429?origin=scopusAI

39. Integrated optimization on train scheduling and preventive maintenance time slots planning Luan, X., Miao, J., Meng, L., (...), Lodewijks, G. Transportation Research Part C: Emerging Technologies, 2017 https://www.scopus.com/pages/publications/85019542417?origin=scopusAI

40. Simultaneous train rerouting and rescheduling on an N-track network: A model reformulation with network-based cumulative flow variables Meng, L., Zhou, X. Transportation Research Part B: Methodological, 2014 https://www.scopus.com/pages/publications/84902346437?origin=scopusAI

41. Planning and scheduling to minimize tardiness Hooker, J.N. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2005 https://www.scopus.com/pages/publications/33646171116?origin=scopusAI

42. Matheuristic strategy development and benchmarking for explainable and efficient aircraft departure sequencing Dönmez, K., Sahin, O. Aeronautical Journal, 2026 https://www.scopus.com/pages/publications/105032239390?origin=scopusAI

43. An appraisal of a column-generation-based algorithm for centralized train-conflict resolution on a metropolitan railway network Min, Y.-H., Park, M.-J., Hong, S.-P., Hong, S.-H. Transportation Research Part B: Methodological, 2011 https://www.scopus.com/pages/publications/78650824016?origin=scopusAI

44. Integrated Reinforcement Learning and Optimization for Railway Timetable Rescheduling Zhang, H., Liu, X., Sun, D., (...), De Schutter, B. IFAC-PapersOnLine, 2024 https://www.scopus.com/pages/publications/85201617839?origin=scopusAI

45. Train-scheduling optimization model for railway networks with multiplatform stations Garrisi, G., Cervelló-

Pastor, C. Sustainability (Switzerland), 2020 https://www.scopus.com/pages/publications/85084253423?origin=scopusAI

46. Real-time Rail traffic Management in Closed-loop: Assessment of Alternative Train Route Selection Sharma, B., Pascariu, B., Pellegrini, P., Rodriguez, J. Transportation Research Procedia, 2026 https://www.scopus.com/pages/publications/105035489483?origin=scopusAI

47. From railway disruptions to recovery: An improved Benders decomposition for the dynamic train timetable rescheduling and rolling stock reassignment Buhigiro, N., Kang, L., Lai, Q., (...), Xu, Q. Computers and Industrial Engineering, 2026 https://www.scopus.com/pages/publications/105026902794?origin=scopusAI

48. Data-driven dispatching rules mining and real-time decision-making methodology in intelligent manufacturing shop floor with uncertainty Zhang, L., Hu, Y., Tang, Q., (...), Li, Z. Sensors, 2021 https://www.scopus.com/pages/publications/85110421328?origin=scopusAI

49. Analyzing Fleet Efficiency and Passenger Delay in Demand-Responsive Transit: A Dual-Model Approach With CVRPTW and TAMOS Al-Nabulsi, D., Oh, J.-S., Kwigizile, V., (...), Jin, H.-M. Journal of Advanced Transportation, 2025 https://www.scopus.com/pages/publications/105013566115?origin=scopusAI

50. A novel movement planner system for dispatching trains Bollapragada, S., Markley, R., Morgan, H., (...), Brantley, J. Interfaces, 2018 https://www.scopus.com/pages/publications/85042494731?origin=scopusAI

51. Application of Intelligent Integration Technology for Automatic Monitoring of Urban Rail Transit Engineering Lu, Y. Lecture Notes in Electrical Engineering, 2024 https://www.scopus.com/pages/publications/85184295968?origin=scopusAI

52. Train dispatching management with data-driven approaches: A comprehensive review and appraisal Wen, C., Huang, P., Li, Z., (...), Xu, X. IEEE Access, 2019 https://www.scopus.com/pages/publications/85079862726?origin=scopusAI

53. Overview of research and development activities relating to digitalization of railway operation management and future outlook Kawasaki, K. Quarterly Report of RTRI (Railway Technical Research Institute), 2020 https://www.scopus.com/pages/publications/85097823103?origin=scopusAI

54. The intelligent dispatching system in the BRT Yao, H., Yu, J. Proceedings of the 8th International Conference of Chinese Logistics and Transportation Professionals - Logistics: The Emerging Frontiers of Transportation and Development in China, 2008 https://www.scopus.com/pages/publications/70249150036?origin=scopusAI

55. Railway Equipment Data Fusion Technology based On "One Map" Liu, Y., Deng, J., Tian, G., Chen, Z. International Conference on Distributed Systems, Computer Networks and Cybersecurity, ICDSCNC 2024, 2024 https://www.scopus.com/pages/publications/105002885114?origin=scopusAI

56. Big data in ITS: Concept, case studies, opportunities, and challenges Moharm, K.I., Zidane, E.F., El-Mahdy, M.M., El-Tantawy, S. IEEE Transactions on Intelligent Transportation Systems, 2019 https://www.scopus.com/pages/publications/85054472573?origin=scopusAI

57. Real-time operations intelligence Kharbat, F.T., Bauer, G. Technical Papers of ISA, 2004 https://www.scopus.com/pages/publications/23844462020?origin=scopusAI

58. Research on Maglev Pipeline Trains Speed and Positioning Based on Improved Kalman Filtering Algorithm Jiang, J., Yang, J., Meng, C., Li, Z. Lecture Notes in Electrical Engineering, 2024 https://www.scopus.com/pages/publications/85186652408?origin=scopusAI

59. A Cross-System Dynamic Data Fusion Method Based on Train Operation Trajectories Zou, D., Sun, S., Wang, P., Wu, J. Lecture Notes in Electrical Engineering, 2025 https://www.scopus.com/pages/publications/105002145296?origin=scopusAI

60. Multi-Source Heterogeneous Data Fusion in Smart Industrial Systems: Intelligent Monitoring, Predictive Quality Assessment, and Fault Diagnosis for Safer Operations Patil, V.B., Patil, S., Patil, S., (...), Charles, P. Journal of Intelligent Decision Making and Information Science, 2026 https://www.scopus.com/pages/publications/105048315030?origin=scopusAI

61. Semantic Technologies for Harmonising European Railway: Improving Interoperability, Data Integration, and Operational Efficiency Sohi, S. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105020239740?origin=scopusAI

62. Optimizing Railway Signaling and Platform Management with LoRaWAN, RFID and Automation Technologies Sarvade, S., Shirwal, V., Kugaonkar, P., Mudgonda, K. Procedia Computer Science, 2025 https://www.scopus.com/pages/publications/85219180036?origin=scopusAI

63. Real-time asynchronous conflict solving algorithms for computer aided train dispatching assistance systems Kuckelberg, A., Wendler, E. WIT Transactions on the Built Environment, 2008 https://www.scopus.com/pages/publications/58849157698?origin=scopusAI

64. Real-time Train Tracking System in Sri Lanka Jayasinghe, U., Kathriarachchi, P. 2021 10th International Conference on Information and Automation for Sustainability, ICIAfS 2021, 2021 https://www.scopus.com/pages/publications/85119966631?origin=scopusAI

65. Real-Time Train Tracking and Time Prediction System For Railway System in Sri Lanka Wanniarachchi, H., Ranasinghe, S., Karunanayake, P., Karunaratne, G. 2024 4th Asian Conference on Innovation in Technology, ASIANCON 2024, 2024 https://www.scopus.com/pages/publications/85218159824?origin=scopusAI

66. Information source quality in intelligent transport systems Jelušić, N., Anžek, M., Ivanković, B. Promet - Traffic and Transportation, 2010 https://www.scopus.com/pages/publications/77952362562?origin=scopusAI

67. Evaluation framework for key performance indicators of railway its Mlinarić, T.J., Đorđević, B., Krmac, E. Promet - Traffic and Transportation, 2018 https://www.scopus.com/pages/publications/85053357443?origin=scopusAI

68. Determining the Significance of Criteria Quality of Passengers Railway Traction Using Expert Evaluation Methods Meilus, L., Maskeliūnaitė, L., Sivilevičius, H. Lecture Notes in Networks and Systems, 2025 https://www.scopus.com/pages/publications/105002575990?origin=scopusAI

69. MILP-Based Optimization of the Extended Real- Time Railway Traffic Management Problem Lindenmaier, L., Aradi, S., Becsi, T., Levetei, I.F. SACI 2022 - IEEE 16th International Symposium on Applied Computational Intelligence and Informatics, Proceedings, 2022 https://www.scopus.com/pages/publications/85141656185?origin=scopusAI

70. Solving the Real-Time Train Dispatching Problem by Column Generation Schälicke, M., Nachtigall, K. Transportation Science, 2025 https://www.scopus.com/pages/publications/105007064639?origin=scopusAI

71. An exact decomposition approach for the real-time train dispatching problem Lamorgese, L., Mannino, C. Operations Research, 2015 https://www.scopus.com/pages/publications/84925357763?origin=scopusAI

72. Multi-Agent Optimization for Railway Network Scheduling and Coordination George, S., Manassa, R.,

Medhat, M., (...), Shehata, O.M. 2026 IEEE International Conference on Smart Sustainable Systems for Computer and Engineering Applications, 3SCEA 2026, 2026 https://www.scopus.com/pages/publications/105046059610?origin=scopusAI

73. Decomposition and distributed optimization of real-time traffic management for large-scale railway networks Luan, X., De Schutter, B., Meng, L., Corman, F. Transportation Research Part B: Methodological, 2020 https://www.scopus.com/pages/publications/85090980408?origin=scopusAI

74. The Component Technology and Evaluation Methods for Railway Signaling Systems in Cloud Platform Zang, Y. Proceedings - 2025 27th IEEE International Conference on High Performance Computing and Communications, 11th IEEE International Conference on Data Science and Systems, 23rd IEEE International Conference on Smart City, 11th IEEE International Conference on Dependability in Sensor, Cloud, and Big Data Systems and Applications and 21st IEEE International Conference on Embedded Software and Systems, HPCC/DSS/SmartCity/DependSys/ICESS 2025, 2025 https://www.scopus.com/pages/publications/105022689325?origin=scopusAI

75. An Optimization-Based Decision Support Tool for Incremental Train Timetabling Kloster, O., Luteberget, B., Mannino, C., Sartor, G. Operations Research Forum, 2023 https://www.scopus.com/pages/publications/85169555317?origin=scopusAI

76. Stability analysis of railway dispatching plans in a stochastic and dynamic environment Quaglietta, E., Corman, F., Goverde, R.M.P. Journal of Rail Transport Planning and Management, 2013 https://www.scopus.com/pages/publications/84901007306?origin=scopusAI

77. Integrating train scheduling and delay management in real-time railway traffic control Corman, F., D'Ariano, A., Marra, A.D., (...), Samà, M. Transportation Research Part E: Logistics and Transportation Review, 2017 https://www.scopus.com/pages/publications/84964589432?origin=scopusAI

78. Analysis of research status and development trends of railway system optimization Liu, P., Zhang, K., Wang, M. Railway Sciences, 2026 https://www.scopus.com/pages/publications/105048041299?origin=scopusAI

79. A Novel Multi-Agent-Based Approach for Train Rescheduling in Large-Scale Railway Networks Liu, J., Chen, L., Tian, Z., (...), Roberts, C. Applied Sciences (Switzerland), 2025 https://www.scopus.com/pages/publications/105011691292?origin=scopusAI

80. Dynamic Scheduling, Operation Control and Their Integration in High-Speed Railways: A Review of Recent Research Dai, X., Zhao, H., Yu, S., (...), Chai, T. IEEE Transactions on Intelligent Transportation Systems, 2022 https://www.scopus.com/pages/publications/85121383172?origin=scopusAI

81. Navigating the Cybersecurity Landscape in Railway Infrastructure: Measures, Challenges, and Innovation Verma, A.K., Lavania, S. Railway Engineering Technology: Developments and Innovations, 2025 https://www.scopus.com/pages/publications/105019154785?origin=scopusAI

82. Rescheduling railway traffic taking into account minimization of passengers' discomfort Corman, F., Pacciarelli, D., D'Ariano, A., Samà, M. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2015 https://www.scopus.com/pages/publications/84951164403?origin=scopusAI

83. Interactions and equilibrium between rescheduling train traffic and routing passengers in microscopic delay management: A game theoretical study Corman, F. Transportation Science, 2020 https://www.scopus.com/pages/publications/85090957128?origin=scopusAI

84. Mining railway traffic control logs Mannhardt, F., Landmark, A.D. Transportation Research Procedia, 2019 https://www.scopus.com/pages/publications/85062220838?origin=scopusAI

85. Revolutionizing railway systems: A systematic review of digital twin technologies Thompson, E.A., Lu, P., Alimo, P.K., (...), Abbew, C.K. High-speed Railway, 2025 https://www.scopus.com/pages/publications/105007556529?origin=scopusAI

86. Microservices for Cloudification and Orchestration of Railway Operations Atanasov, I., Pencheva, E., Trifonov, V. Communications in Computer and Information Science, 2025 https://www.scopus.com/pages/publications/85208227774?origin=scopusAI

87. Continuous-Space Multi-Agent Path Finding via Enhanced Prioritized Search with Ackermann Kinematic Constraints Zhang, T., Zhang, L., Cai, Q., (...), Wang, J. IEEE Robotics and Automation Letters, 2026 https://www.scopus.com/pages/publications/105035306980?origin=scopusAI

88. Advancing Learnable Multi-Agent Pathfinding Solvers with Active Fine-Tuning Andreychuk, A., Yakovlev, K., Panov, A., Skrynnik, A. IEEE International Conference on Intelligent Robots and Systems , 2025 https://www.scopus.com/pages/publications/105029916910?origin=scopusAI

89. When to Switch: Planning and Learning for Partially Observable Multi-Agent Pathfinding Skrynnik, A., Andreychuk, A., Yakovlev, K., Panov, A.I. IEEE Transactions on Neural Networks and Learning Systems, 2024 https://www.scopus.com/pages/publications/85169705242?origin=scopusAI

90. Multi-Agent Path Finding Based on Graph Neural Network Li, X., Gao, J., Li, Y. Chinese Control Conference, CCC, 2023 https://www.scopus.com/pages/publications/85175526973?origin=scopusAI

91. SCRIMP: Scalable Communication for Reinforcement- and Imitation-Learning-Based Multi-Agent Pathfinding Wang, Y., Xiang, B., Huang, S., Sartoretti, G. IEEE International Conference on Intelligent Robots and Systems, 2023 https://www.scopus.com/pages/publications/85182523007?origin=scopusAI

92. PRIMAL₂: Pathfinding Via Reinforcement and Imitation Multi-Agent Learning-Lifelong Damani, M., Luo, Z., Wenzel, E., Sartoretti, G. IEEE Robotics and Automation Letters, 2021 https://www.scopus.com/pages/publications/85102262872?origin=scopusAI

93. A Domain-Specific Question-Answering System for Railway Data: A Hybrid Approach Both, A., Herbst, K., Krieg, R., (...), Diettrich, A. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105020246038?origin=scopusAI